Learning Objectives

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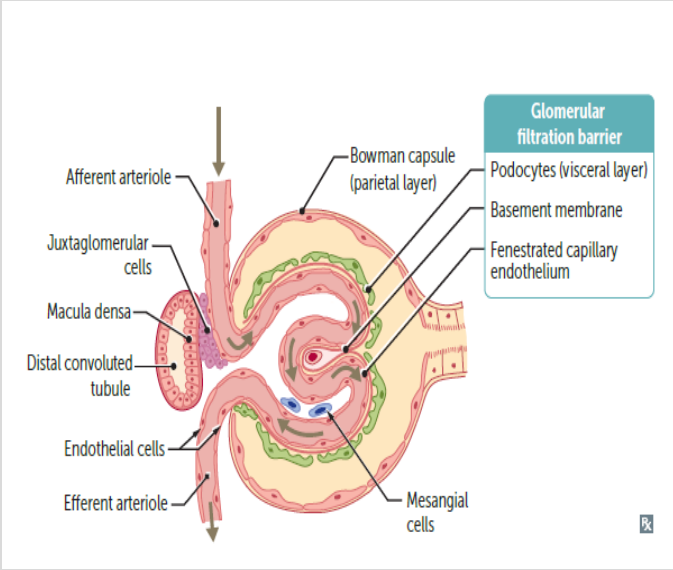
Learning Objectives By the end of this introductory session, students should be able to:

1. Describe the basic structure and function of the glomerulus within the nephron
2. Explain how disruptions in the glomerular filtration barrier lead to clinical disease
3. Understand general mechanisms that contribute to glomerular injury (immune complex deposition, antibody- and cell-mediated damage, complement activation)
4. Differentiate broadly between nephrotic and nephritic syndromes in terms of pathophysiology and clinical features
5. Appreciate the role of diagnostic tools such as light microscopy, immunofluorescence, and electron microscopy in evaluating glomerular pathology
6. Recognize hallmark features seen in common glomerular diseases without going into subtype-level details

The Glomerular Filtration Barrier

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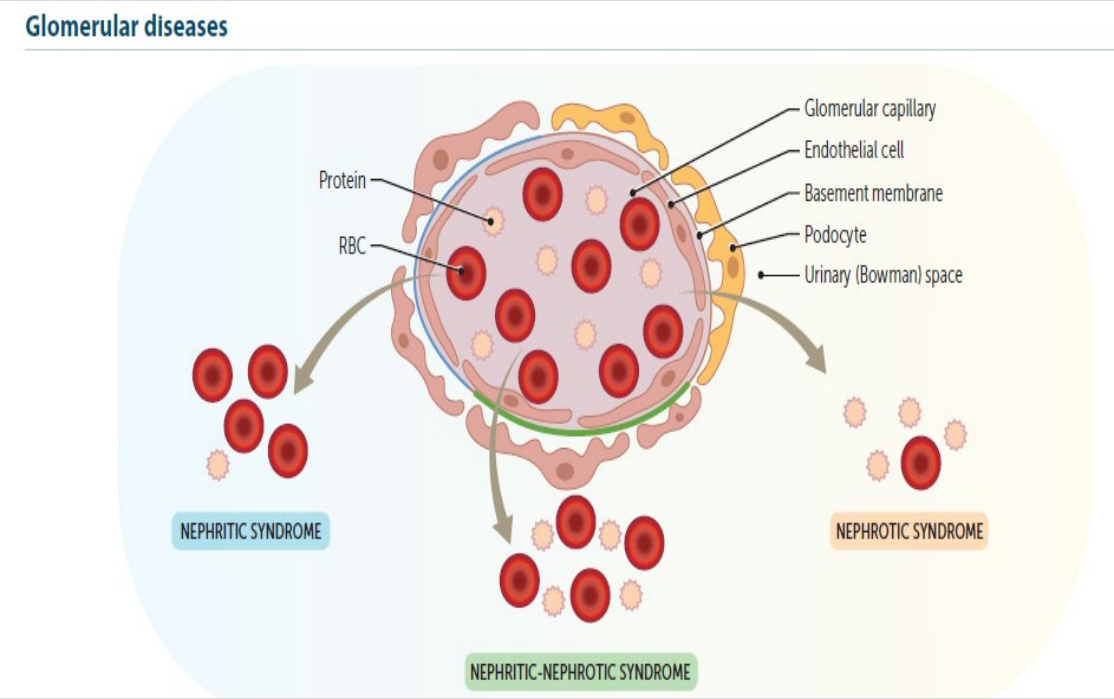
- Fenestrated capillary endothelium :
  - Fenestrated (~70nm)
  - Allows water, solutes, and proteins
- Glomerular basement membrane (GBM) - negatively charged:
  - Repels negatively charged molecules, e.g. proteins



First Aid for the USMLE Step 1 2023

Glomerular Diseases presentation

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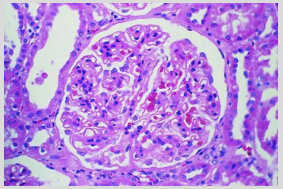
- Damage to glomerular morphology causing disruption of the filtration barrier
- Can cause leakage of either:
  - Protein in urine (proteinuria)
  - Blood in urine (Hematuria)

Diagnostic Techniques for Glomerular Disease:

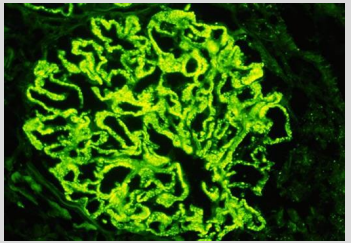
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Diagnosis is guided by clinical presentation and confirmed through renal biopsy analysis

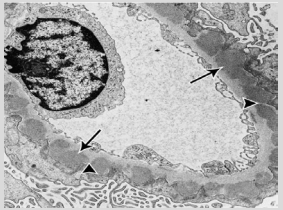
- Light microscopy (LM)
  Gross structure of glomerulus and cellularity



- Immunofluorescence (IF)
  Pattern and type of protein (Ab, complement) deposition



- Electron microscopy (EM)
  Identifies location of protein deposition
  Foot process effacement



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Light Microscopy (LM)

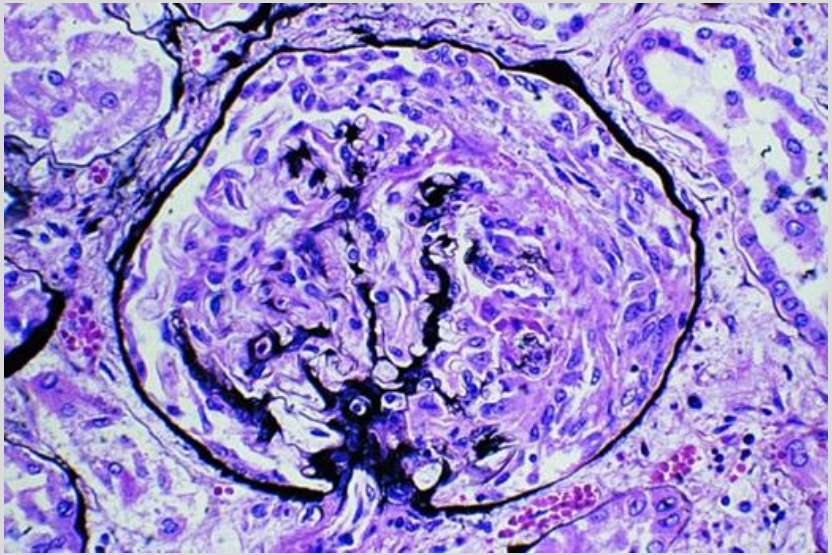
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- **Identifies areas of**
  - Increased cellularity
  - Increased extracellular matrix
- **Stains**
  - H&E: basic morphology
  - PAS: basement membrane & mesangium
  - Silver stain: GBM pattern
  - Congo Red: amyloid
  - Trichrome: fibrosis, sclerosis

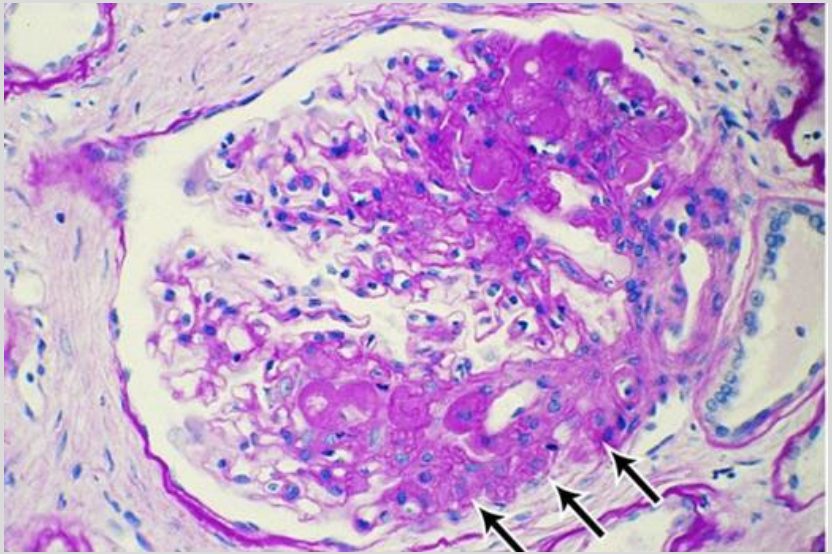
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Light Microscopy

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Silver staining (shows ANCA glomerulonephritis)

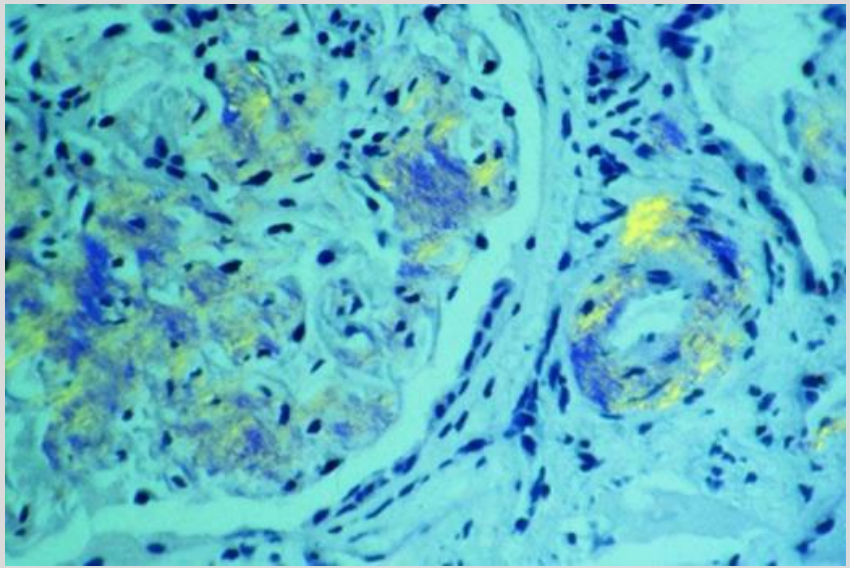


PAS staining (shows FSG)

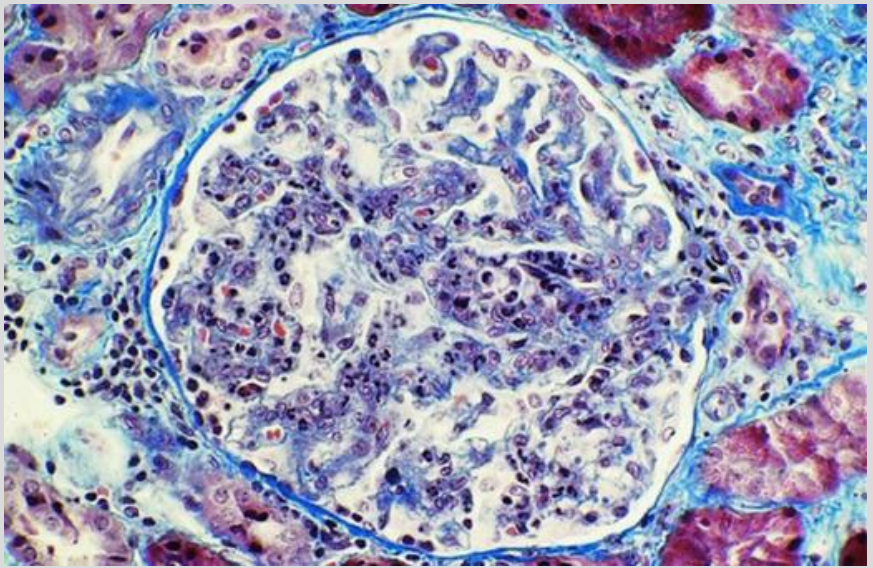
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Light Microscopy

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Congo red staining (shows amyloid nephropathy)



Trichrome staining (shows acute PSGN)

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Immunofluorescence

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- **Identifies patterns of:**

Linear: anti-GBM (Goodpasture)

Granular: immune complexes (PSGN, lupus)

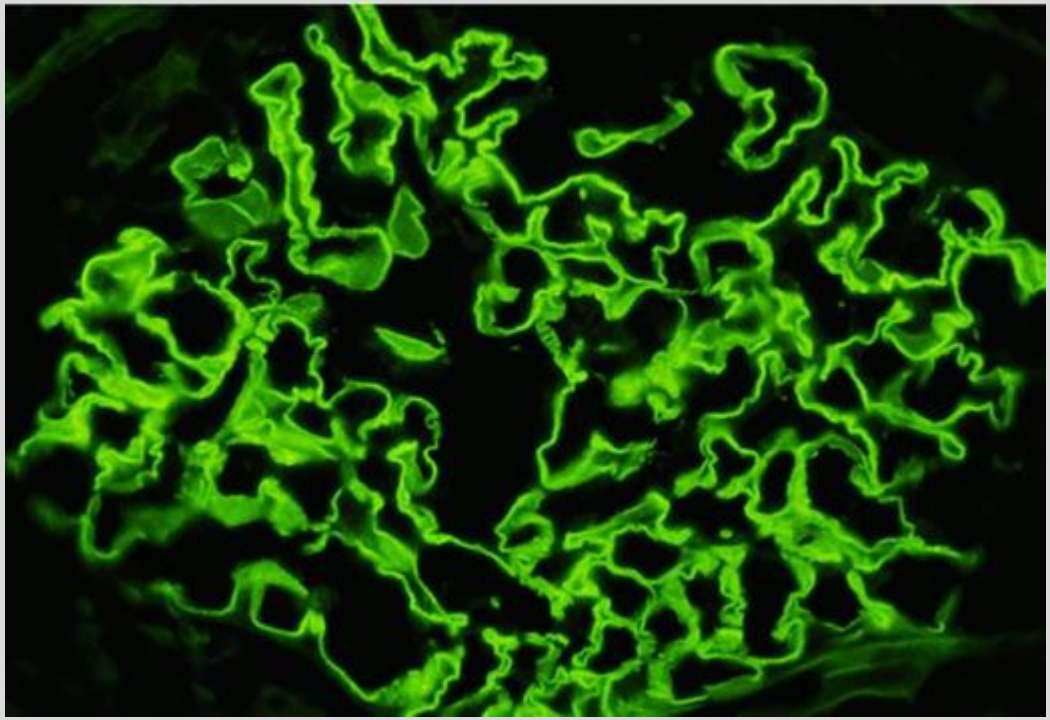
Irregular/amorphous: IgA nephropathy

- **Detects:**
  - Albumin
  - Fibrinogen
  - Antibodies (IgA, IgG, IgM)
  - Complement (C1q, C3, C4)

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Linear Immunofluorescence Staining

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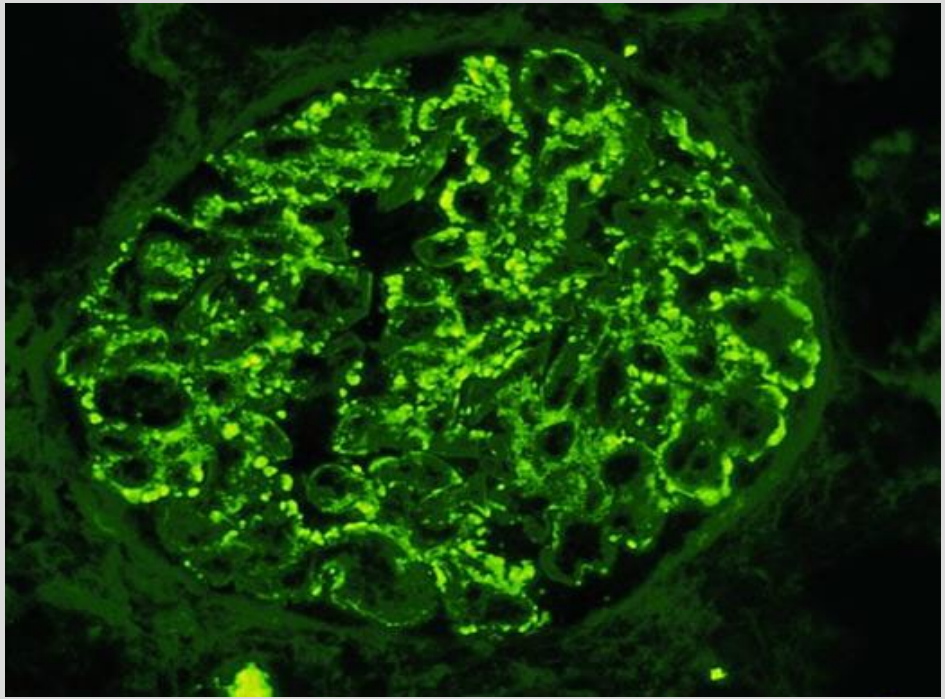


- Immunostaining for IgG
- Diffuse linear pattern
- Seen in anti-GBM disease
  - Type 2 hypersensitivity reaction

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Granular Immunofluorescence Staining

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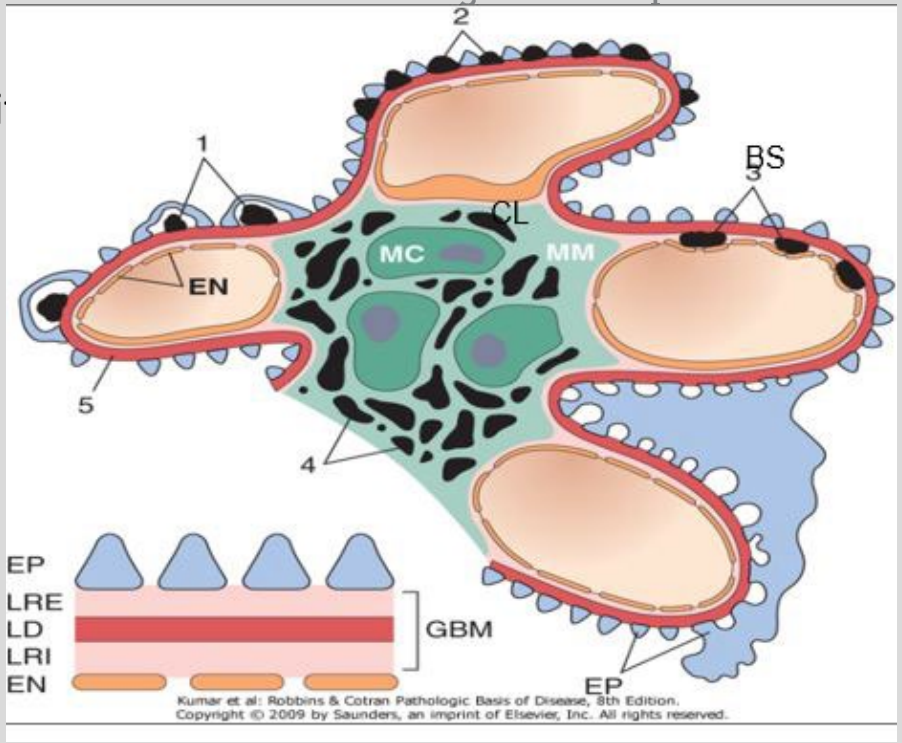
- Immunostaining for immunocomplex deposits
- Diffuse granular pattern
  - "lumpy-bumpy"
- Seen in acute post-streptococcal glomerulonephritis
  - Type 3 hypersensitivity reaction

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Electron Microscopy

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- Identifies:
  - Electron dense immune complex deposi
    - Subepithelial humps/deposits
    - Subendothelial
    - Mesangial
  - GBM thickening and remodeling
  - Collagenous matrix
  - Fibrillar deposits
- Detects:
  - Effacement of foot processes
  - Endothelial injury

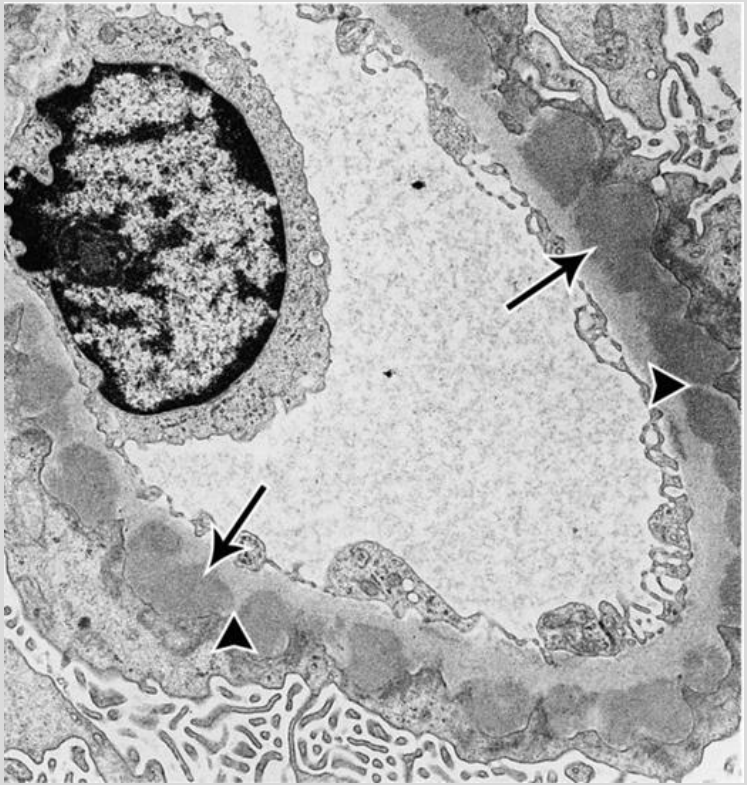


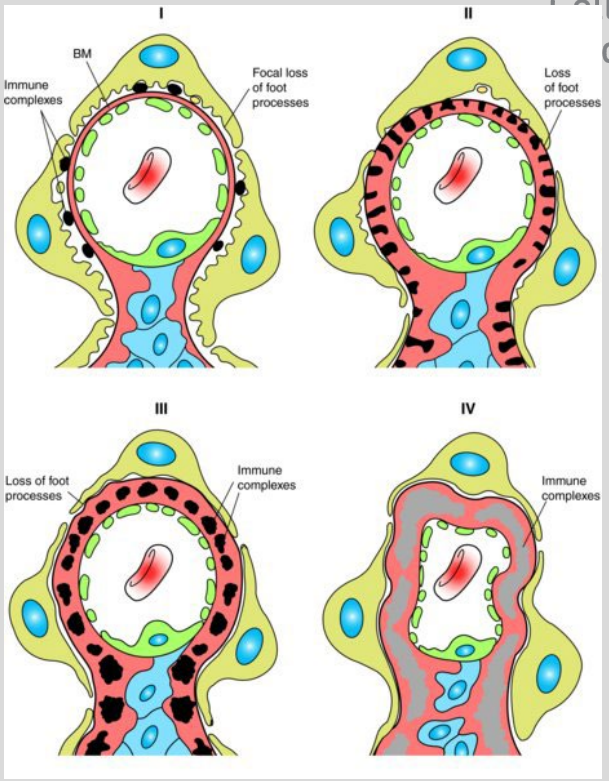
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Vinay, Kumar. *Robbins & Cotran pathological Basis of Disease 8e*

Electron Microscopy (Subepithelial)

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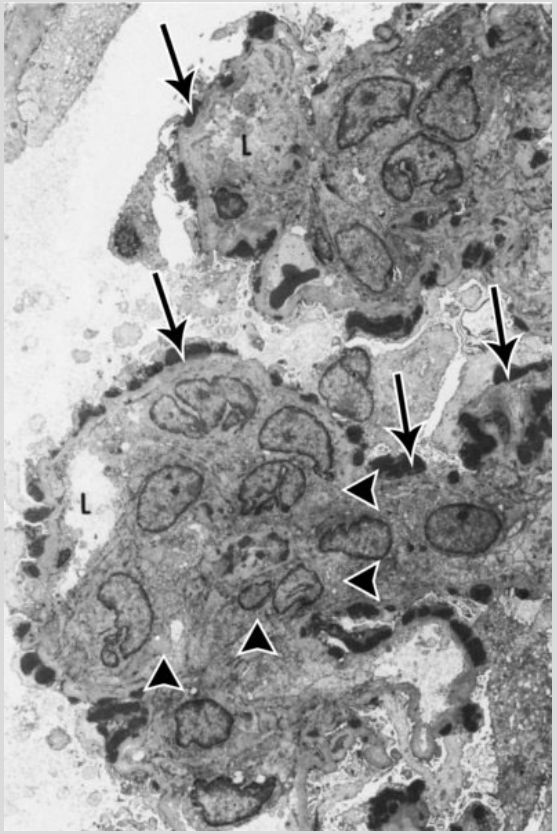


- Membranous nephropathy

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Electron Microscopy (Subepithelial)

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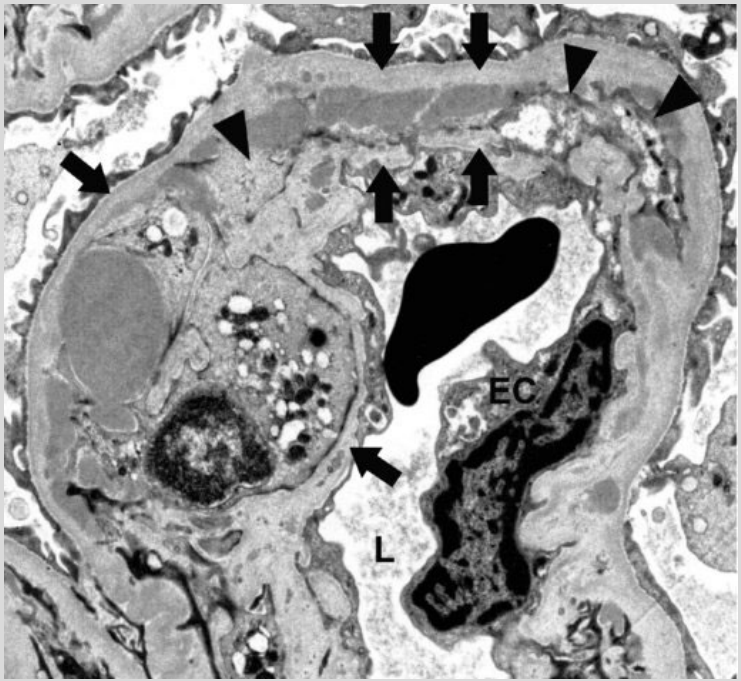


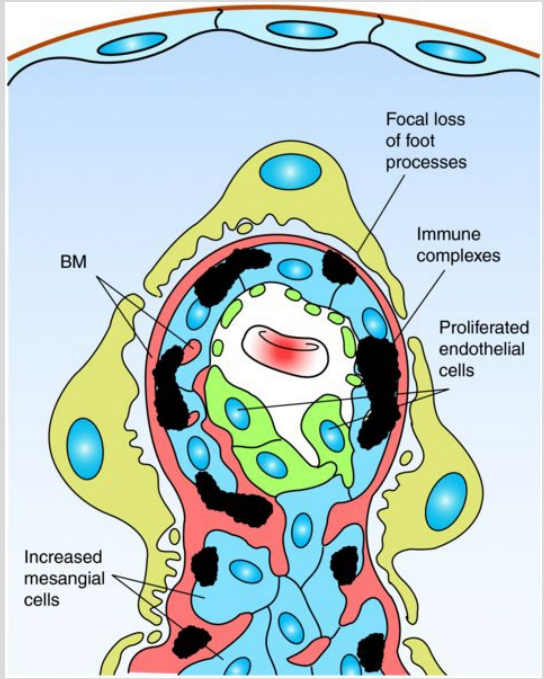
- Acute post-streptococcal glomerulonephritis (PSGN)
- Subepithelial humps
- Mesangial hypercellularity

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Electron Microscopy (Subendothelial)

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- Membranoproliferative glomerulonephritis
  - "Tram track" appearance
- Subendothelial deposits with mesangial cell proliferation

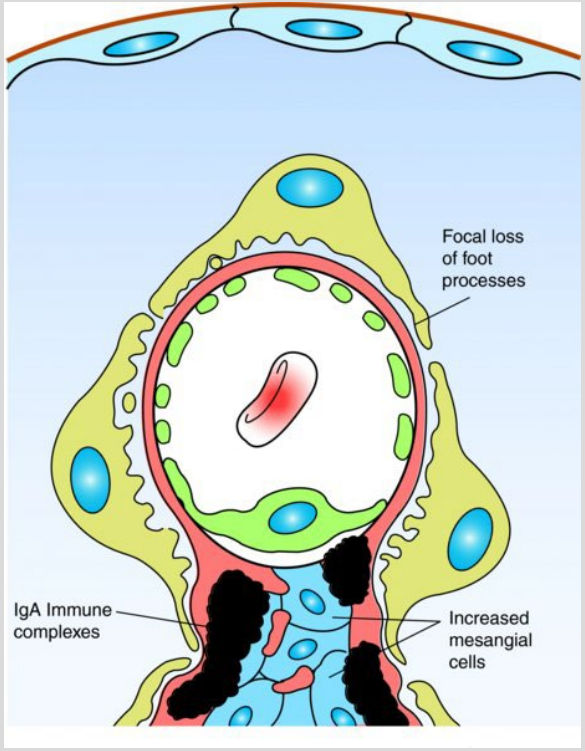
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Electron Microscopy (Mesangial)

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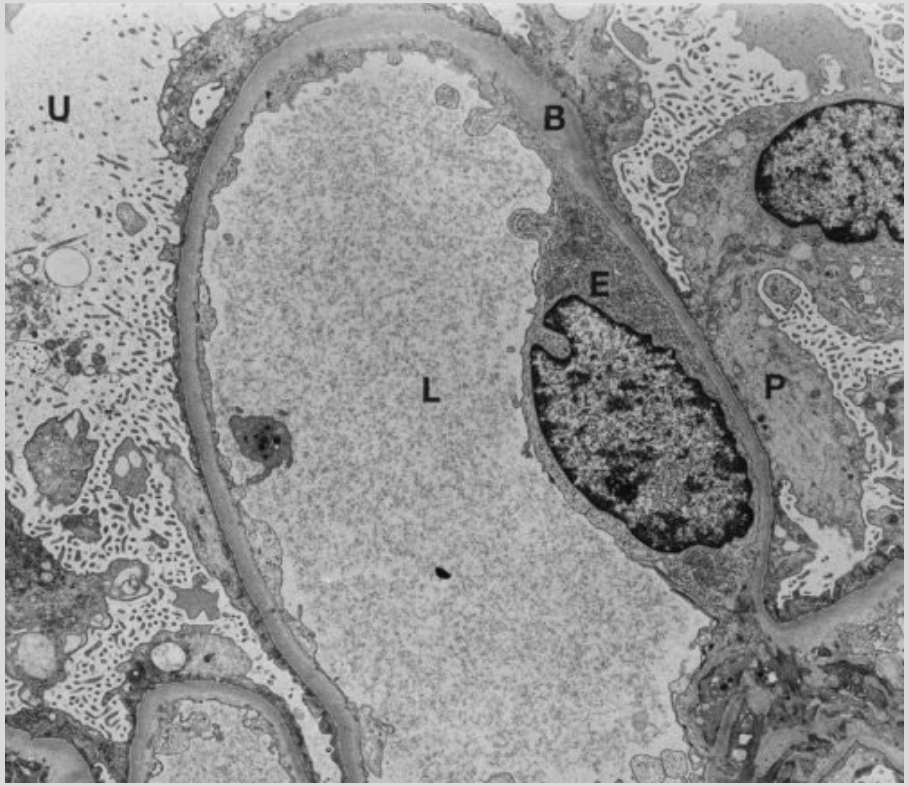


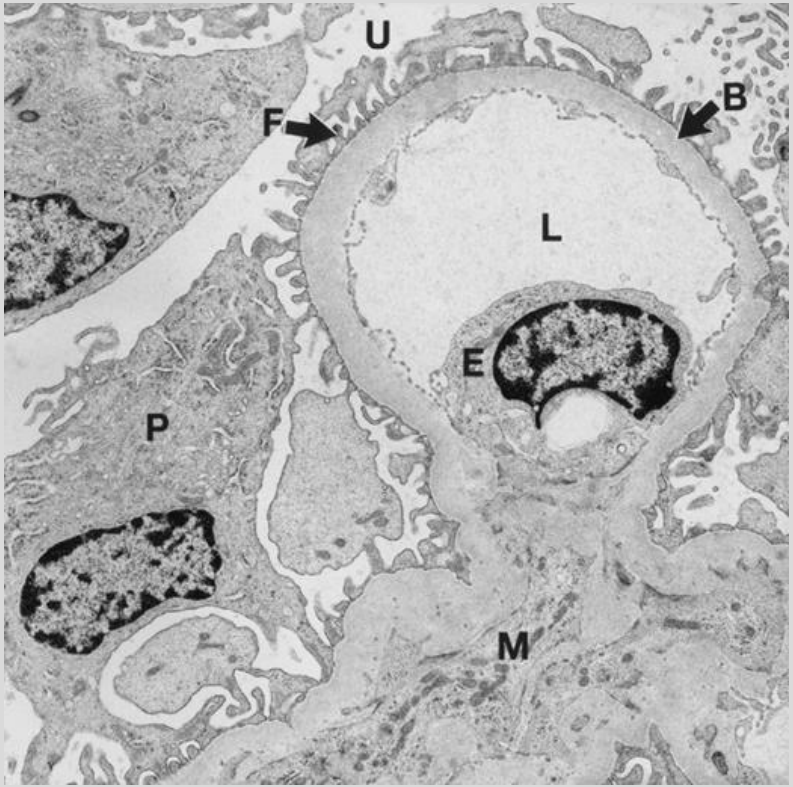
- IgA nephropathy
- Mesangial matrix deposits between mesangial cells and GBM

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Effacement of the foot processes

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Summary

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- Filtration barrier: critical to glomerular function
- Glomerular disease = damage to barrier
- Nephrotic syndrome: characterized by proteinurea and hypoalbuminemia (PALE)
- Nephritic syndrome: characterized by inflammation of the glomeruli and resulting hematuria
- Microscopy essential for diagnosis

Sample Question

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- Which of the following findings best differentiates nephritic from nephrotic syndrome on urinalysis?

  A) Presence of hyaline casts
  B) Presence of dysmorphic RBCs and RBC casts
  C) Presence of fatty casts
  D) Low urine sodium

Explanation:
Nephritic syndrome involves glomerular inflammation, leading to **hematuria** with **dysmorphic red blood cells** and **RBC casts** on urinalysis. This finding is **specific to nephritic syndrome** and helps differentiate it from nephrotic syndrome, which is marked by heavy proteinuria and **fatty casts**.